

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	NSE
People present	72, APA, BAV, Josie Lindy-Clarke, Eggert, Björn
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	21/5-2024

Notes:

no snowboard visible from 2023
→ new snowboard placed 10m due N from AWS mast

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y		212890	244328	Clean? Y / N slightly rimed
Iridium modem IMEI number	N	30053	4063818800		
Inclinometer/compass (Inside CNR4)	Y	Y	Attached to the radiometer? Y / N		
Satellite antenna	N	N			
Wind sensor upper	Y	N	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	Y	Y	Black box pointing same direction as the radiometer? Y / N		
Temperature / humidity assembly upper	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Temperature / humidity assembly lower	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper	Y	Y	Old sensor with new membrane? Y / N		
Sonic ranger lower	Y	Y	Old sensor with new membrane? Y / N		
Thermistor string	Y	new setup	147 cm tube out of snow		
Solar panel	N	N	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N	N	At top of logger enclosure? Y / N		
Logger box ID	N	M0003 M0003	CR1000 or CR1000X? ID number:		
Logger enclosure	N	N	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	Y	New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	Y	Y	How many inserted? How long? Notes: 1 150 + 0,8 cm		
Mast	N	N	New mast? Y / N New sensor boom? Y / N		

AKSS dist. from bottom of mount
to mast connection: 18,3 cm

Station name	NSE <i>37, APA, 3AV, 7051e</i>
Date	21/5-2024

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+ 00:25	+ 00:25
Download data to PC and/or change SD card (wait for green light)	Y ☒ N	Y / N
Name of logger program	no program	GC-11 v. 2.6
Screen dump of values in fast scan mode	☒ / N	☒ / N
Photos of mast, sensors, logger box wiring etc. (more is better)	☒ / N	Y / N
Mast tilt towards radiometer (- if looking up)	-3°	-4°
Mast tilt across boom looking from radiometer (- if clockwise)	-2°	-2°
Upper boom direction with radiometer relative to north	0° true/magn.	0° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	☒ / N °	☒ / N °
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	☒ / N °	☒ / N °
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	☒ / N °	☒ / N °
Upper temperature / humidity height (bottom of casing – surface)	258 cm	~ 410 cm
Lower temperature / humidity height (bottom of casing – surface)	134 cm	290 cm
Upper sonic ranger height (bottom of casing – surface)	199 cm	~ 346 cm
Lower sonic ranger height (bottom of casing – surface)	83 cm	232 cm
Length of thermistor ^{tube height} string on surface from surface marking	N/A cm	147 cm
Apparent damage to sensors? If yes, which?	☒ / N	Y / ☒ N

Notes:

! Antenna cable disconnected from logger box, and lying buried in snow. No sign of damage to the plug. Likely not properly screwed / fastened (possibly cross-threaded) program still on cpu
more than 11.000 watch dog errors (not warnings)
program errors: 1
battery voltage: 14.68 V
lithium battery: 3.59 V
online prop-head of Yang sensor missing

Must-do at next visit:

below 12V > 263 mV then
c1 bus error → ~~prob~~ speculating "no program" issue on Luftts
no data on logged cpu
→ saw the same on G105 stations
OS version 0.6.00 (needs upgrade)
changed SD card
→ upgraded to OS 06.02

running in fastscan mode very slow and does not count down → seems to be solved by disconnecting the two Luftt instrum.
disconnecting solar panel → battery voltage still 14.5V